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An explainable machine learning framework for analyzing and predicting mental health problems among university students in Bangladesh

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Mental health problems (MHPs) among university students are an increasing public health concern globally, including in Bangladesh. While machine learning (ML) methods can capture complex patterns in mental health data, their application to non-probability, cross-sectional student samples requires cautious interpretation. This study applies an exploratory explainable ML framework to examine patterns of depression, anxiety, and stress within a high-symptom sample of Bangladeshi university students. A cross-sectional online survey was conducted among 424 students during a period of significant social and political unrest in July 2024. Within the study sample, high proportions of moderate-to-extremely severe symptoms were observed for depression (66.0%), anxiety (73.4%), and stress (52.4%), with substantial symptom overlap (48.1% meeting criteria for all three). Eight ML models were evaluated using repeated train–test splits and cross-validation. Support vector machine (SVM) models showed relatively stable performance for depression and stress prediction, while XGBoost demonstrated comparatively stronger discriminative ability for anxiety within the sample. Explainable ML analyses using SHAP and LIME indicated that features related to sleep disturbance, cognitive complaints, and self-reported behavioral difficulties contributed prominently to model predictions across outcomes. Academic discipline and internet addiction severity were more influential in anxiety-related predictions, whereas gender and university type showed greater relevance in depression-related models. The primary contribution of this study is methodological and exploratory, not predictive or clinical. These feature contributions reflect model behavior and overlapping symptom dimensions rather than independent or causal risk factors. Overall, this study demonstrates the feasibility of using explainable ML approaches for exploratory pattern recognition in student mental health data under exceptional contextual conditions, providing a transparent framework, benchmark performance metrics, and a methodological template for future research. The findings are exploratory and sample-specific and should not be interpreted as population-level prevalence estimates or deployable prediction tools.

Keywords Explainable machine learning, Local interpretable model-agnostic explanations, Mental health, SHapley Additive exPlanations, K-fold cross-validation

Mental health problems (MHPs), including depression, anxiety, and stress, are among the leading contributors to disability and reduced quality of life worldwide¹. According to the Global Burden of Disease (GBD) Study 2021, anxiety and depressive disorders affected approximately 359.2 million and 332.4 million people globally, respectively, accounting for 9.1% of all diseases and 63.1% of the total mental disorder burden². Mental health

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refers broadly to an individual's emotional, psychological, and social well-being, whereas mental illness denotes clinically diagnosable conditions characterized by disturbances in mood, cognition, or behavior^{1,3}. According to the World Health Organization (WHO), mental disorders are common across the life course and can substantially impair daily functioning if left unaddressed¹. Among young adults, untreated mental health problems are associated with adverse outcomes such as impaired academic performance, social withdrawal, and elevated risk of self-harm and suicide^{4,5}.

University students represent a population particularly vulnerable to mental health difficulties due to academic pressure, financial uncertainty, transitional life stages, and evolving social roles^{6,7}. A large body of empirical research conducted across diverse countries has consistently documented high levels of depressive, anxiety, and stress symptoms among university students^{6–12}. These studies highlight that mental health concerns are not confined to a single cultural or economic context but represent a global challenge within higher education systems. Importantly, most estimates are derived from self-reported screening tools and reflect symptom burden within study samples rather than population-level prevalence.

In Bangladesh, mental health problems among university students have attracted increasing research attention. Prior studies have reported substantial levels of depression, anxiety, and stress among medical and non-medical students using validated screening instruments such as the DASS-21^{13–17}. Nationally representative evidence underscores the public health significance of mental disorders in the general population: the Bangladesh National Mental Health Survey 2019 found that nearly 18.7% of adults and 12.6% of children experience a diagnosable mental disorder, with depressive disorders (6.7%) and anxiety disorders (4.7%) among the most common conditions reported¹⁸. A systematic review of mental disorders in Bangladesh further underscores the growing public health relevance of mental health concerns in the country, particularly among young people¹⁷. These findings suggest that Bangladeshi university students may face a considerable mental health burden, although estimates vary widely depending on study design, sampling strategy, and contextual factors.

Mental health problems among young adults have also been linked to serious health consequences beyond psychological distress. Evidence indicates that mental disorders are associated with increased risk of self-harm and suicide⁴, as well as elevated cardiovascular risk, including heart attack and stroke, in young adult populations^{19,20}. While such outcomes are not universal, they emphasize the importance of early detection, monitoring, and appropriate support for mental health problems during early adulthood.

In recent years, machine learning (ML) methods have been increasingly applied to mental health research due to their capacity to handle high-dimensional data, capture complex non-linear relationships, and integrate diverse behavioral and psychosocial variables^{21,22}. Prior studies have demonstrated the feasibility of ML approaches for predicting mental health outcomes using survey data, electronic health records, and behavioral indicators^{23–26}. In the context of university students, ML techniques such as logistic regression, support vector machines, random forests, and gradient boosting models have been employed to predict depression, anxiety, stress, and related outcomes with varying levels of performance^{26–29}.

In Bangladesh, a limited number of studies have applied ML to student mental health data. Existing work has focused on specific outcomes such as perceived stress²⁷, depression²⁸, or anxiety and insomnia²⁹, often using binary classification frameworks. While these studies demonstrate the technical feasibility of ML approaches, they typically emphasize predictive accuracy without sufficient attention to interpretability or contextual limitations. Moreover, many ML models function as “black boxes,” making it difficult to understand which features contribute most strongly to predictions and how these features relate to students' lived experiences. Explainable machine learning (XAI) techniques offer a potential solution to this challenge by providing transparent insights into model behavior and feature contributions. Methods such as SHAP and LIME allow researchers to examine how specific variables influence model predictions at both global and individual levels, without implying causal relationships^{30–32}. In mental health research, such interpretability is essential for ensuring responsible use of ML models, particularly when outcomes are sensitive and potentially stigmatizing.

Against this background, the present study applies an explainable machine learning framework to examine patterns of depression, anxiety, and stress among university students in Bangladesh. Using data collected through a cross-sectional online survey and employing eight commonly applied ML algorithms, including decision trees, random forests, logistic regression, support vector machines, AdaBoost, XGBoost, LightGBM, and artificial neural networks, we evaluate model performance and explore feature contributions using explainability techniques. The goal of this study is not to infer causality or estimate population-level prevalence, but rather to investigate symptom patterns, model behavior, and feature associations within a self-selected student sample. By explicitly acknowledging methodological constraints, this work seeks to contribute to the growing literature on the exploratory use of explainable ML methods in student mental health research in low- and middle-income country settings.

Materials and methods

Participants and procedure

Proper sampling methodology is important in mental health research to ensure valid and reliable inference. In the present study, a non-probability purposive sampling strategy was employed, which is commonly used in online mental health research when probability sampling is not feasible³³. A cross-sectional online survey was conducted between July 1 and 22, 2024, involving university students from various institutions across Bangladesh. Participation was voluntary and anonymous, and data were collected using a structured Google Form. A total of 424 students (290 men and 134 women) completed the questionnaire. As participation was self-selected and based on purposive sampling, the sample does not constitute a representative sample of all Bangladeshi university students and may overrepresent individuals experiencing psychological distress. Consequently, the findings should be interpreted as reflecting patterns within a high-risk, self-selected student sample rather than population-level estimates. No incomplete questionnaires were recorded.

Behavioral and demographic factors

The exploratory variables included a range of demographic, behavioral, and self-reported psychosocial characteristics, such as gender, body mass index (BMI), university type, academic discipline, education level, career aspirations, use of Google Classroom, marital status, participation in physical activity, household chores, smoking status, employment status, number of children, communication patterns, religious activity, dietary habits, vitamin supplementation, and profession. Several variables (e.g., memory loss symptoms, waking from bad dreams, misbehavior with family members) may reflect correlates or manifestations of psychological distress rather than independent antecedent risk factors, and were therefore interpreted cautiously. BMI was categorized following the World Health Organization (WHO, 2010) classification: underweight (BMI < 18.5 kg/m²), normal weight (18.5–24.9 kg/m²), and overweight/obese (BMI ≥ 25 kg/m²)³⁴.

Association analyses

To examine relationships between categorical socio-demographic, behavioral, and psychosocial variables and mental health outcomes (depression, anxiety, and stress), chi-square tests of independence were conducted. For variables with small expected cell counts (<5), Fisher's exact test was applied to ensure valid inference. These analyses were exploratory and conducted to identify potential correlates for subsequent feature selection in machine learning models.

Internet addiction test (IAT)

Internet addiction was assessed using the Internet Addiction Test (IAT) developed by Young³⁵. The instrument consists of 20 items rated on a 5-point Likert scale ranging from “not applicable” (1) to “always” (5), yielding total scores from 20 to 100. Participants were classified as having normal internet use (≤ 30), mild addiction (31–49), moderate addiction (50–79), or severe addiction (≥ 80). For the purposes of this analysis, IAT scores were dichotomized into severe internet addiction (≥ 80) and non-severe internet use (< 80) to focus specifically on severe forms of problematic use relevant for mental health screening. This dichotomization entails information loss and was therefore interpreted cautiously. The Cronbach's alpha for the IAT in this study was 0.919, indicating excellent internal consistency.

Depression, anxiety, and stress scale (DASS-21)

Depression, anxiety, and stress symptoms were assessed using the 21-item Depression, Anxiety, and Stress Scale (DASS-21). The scale comprises three subscales—depression, anxiety, and stress—each consisting of seven items. Responses were recorded on a 4-point Likert scale ranging from 0 (“never”) to 3 (“almost always”), with subscale scores multiplied by two to obtain final scores. Severity classifications followed standard cutoffs: depression (normal 0–9, mild 10–13, moderate 14–20, severe 21–27, extremely severe ≥ 28), anxiety (normal 0–7, mild 8–9, moderate 10–14, severe 15–19, extremely severe ≥ 20), and stress (normal 0–14, mild 15–18, moderate 19–25, severe 26–33, extremely severe ≥ 34)³⁶.

For machine learning modeling, outcomes were dichotomized into elevated symptom levels (“moderate to extremely severe”) versus lower symptom levels (“normal to mild”) using established clinical cut-offs from the DASS-21 manual, resulting in binary target variables (Yes = 1, No = 0). This approach facilitates binary classification modeling and aligns with clinical screening frameworks. We acknowledge that dichotomization reduces information content; however, supplementary analyses treating outcomes as continuous (Supplementary Table S1) yielded similarly modest predictive performance ($R^2 < 0.25$), indicating that the limited predictability reflects underlying data characteristics rather than the binary transformation. The DASS-21 is a well-accepted measure of psychometric qualities with confirmed validity and reliability in non-clinical samples³⁶. The DASS-21 demonstrated high internal consistency in this study (overall Cronbach's alpha = 0.947), with subscale alphas of 0.862 (stress), 0.848 (anxiety), and 0.884 (depression), indicating strong reliability.

Statistical analyses

The study aimed to explore predictive patterns of mental health problems and identify influential correlates using multiple machine learning (ML) models, including Decision Trees (DT), Random Forests (RF), Support Vector Machines (SVM), Logistic Regression (LR), Adaptive Boosting (AdaBoost), Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), and Artificial Neural Networks (ANN). The analytical workflow consisted of the following steps: data preprocessing; splitting the dataset into training (80%) and test (20%) subsets; performing feature selection using the Boruta algorithm exclusively on the training data; training ML models using the selected features; and evaluating model performance on the held-out test set. The best-performing model for each outcome was identified using a multi-criteria evaluation framework. Primary emphasis was placed on the area under the ROC curve (AUC) to assess overall discriminative ability, while sensitivity and specificity were jointly considered to avoid trivial or highly imbalanced classification behavior. In addition, consistency across repeated cross-validation schemes was used as a secondary criterion to prioritize models demonstrating stable performance rather than isolated metric optimization. All analyses were conducted using Python version 3.13.3. Given the purposive, self-selected sampling strategy, analyses were designed to examine associative and predictive patterns within the observed study sample, and results were not intended to support population-level inference.

Boruta algorithm

Key predictive features were identified using the Boruta algorithm, a wrapper method built around random forest classification³⁷. Boruta compares original features with randomized shadow features to determine statistically significant importance. Feature selection was performed separately for each outcome variable using only the training dataset to avoid information leakage.

Decision tree (DT)

The most popular ML technique for creating prediction algorithms for a given variable is DT^{38–44}. The population is segmented into branch-like sections in this instance, which resemble an inverted tree with branches, leaf nodes, and roots^{39,40,44}. While leaf nodes show the final result and branches indicate decision rules, internal nodes hold data characteristics and properties^{39,40,45}.

Random forest (RF)

An RF algorithm has hyperparameters that define the number of trees and the maximum depth of each tree, determining how many interactions are considered in the model. The parameters of the model are the decision rules themselves^{41,46}. RF, an ensemble learning technique for classification, leverages a large set of uncorrelated decision trees (DTs). In our Python implementation of the RF method, we used 1,000 decision trees and applied the Gini impurity index to measure the quality of splits.

Logistic regression (LR)

LR is a probabilistic statistical model used to predict the likelihood of an event occurring^{42,43,47}. It models the relationship between a binary or categorical outcome and one or more independent variables. LR is commonly applied to predict binary or multiple outcomes based on the values of the independent variables, making it suitable for situations where the response variable is categorical^{42,43,48}.

Support vector machine (SVM)

SVM is a type of supervised learning algorithm used to analyze data and identify patterns^{44–47}. For a binary classification task, an SVM training algorithm creates a model or classification function that assigns new observations to one of two classes based on their position relative to a hyperplane, effectively acting as a non-probabilistic linear classifier. SVMs use the kernel trick to map data into a higher-dimensional space, allowing them to solve the machine-learning task as a convex optimization problem^{44–51}. The data points that are closest to the hyperplane, which plays a critical role in defining the boundary, are called support vectors^{44–47,50}. In our analysis, we used SVM with various kernels to evaluate different SVM models.

Adaptive boosting (AdaBoost)

AdaBoost is an ensemble method that combines multiple weak classifiers to create a robust predictive model. Introduced by^{48–52}, AdaBoost trains weak learners, commonly decision stumps, on re-weighted data where misclassified instances are given higher importance in subsequent rounds. This iterative re-weighting helps the model focus on difficult-to-classify examples. The final prediction is based on a weighted majority vote of all weak classifiers, with weights proportional to each classifier's accuracy. AdaBoost effectively improves accuracy and reduces bias in classification tasks, but it can be sensitive to noisy data and outliers^{48–53}.

Extreme gradient boosting (XGBoost)

XGBoost is a highly effective and powerful implementation of gradient boosting designed for supervised learning tasks, including classification and regression. Gradient boosting builds an ensemble of decision trees sequentially, with each tree correcting the errors made by its predecessors by minimizing a specific loss function^{48,51,54}. XGBoost enhances traditional gradient boosting by incorporating system and algorithmic optimizations, adding regularization to prevent overfitting, supporting parallel processing, and automatically handling missing data. It employs a second-order Taylor expansion of the loss function, which improves both convergence speed and model accuracy. Hyperparameters such as tree depth, learning rate, and regularization terms are tuned to optimize predictive performance^{48,51,55}.

Light gradient boosting machine (LightGBM)

Developed by Microsoft, LightGBM is a high-performance gradient boosting framework optimized for large-scale machine learning projects. Like other boosting methods, it builds decision trees iteratively to optimize a loss function via gradient descent. LightGBM introduces two significant innovations: Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB). These techniques reduce the amount of data processed and decrease memory usage without compromising accuracy. Unlike traditional methods that grow trees level-wise, LightGBM grows trees leaf-wise (branch by branch), leading to higher accuracy. Its support for parallel and GPU-based learning makes it suitable for very large datasets. Key hyperparameters such as the number of leaves, learning rate, and boosting iterations are carefully tuned for best performance^{48,52,56}.

Artificial neural networks (ANNs)

ANNs are computational models inspired by the structure of biological neural networks in the human brain. They consist of layers of interconnected nodes, or neurons, organized into input, hidden, and output layers^{53–55,57,58}. Connections between nodes have associated weights that are adjusted during training to minimize the error between predicted and actual outputs. Data propagates through the network via activation functions applied layer by layer, and errors are backpropagated using optimization algorithms such as gradient descent^{53–55,57,58}. ANNs excel at modeling complex nonlinear relationships, making them suitable for various classification and regression applications across diverse domains^{53–55,57,58}.

Confusion matrix performance parameters

A confusion matrix visually represents the accuracy of actual versus predicted classifications^{48,55,56}. It displays the performance of a classification algorithm using true positive, false positive, true negative, and false negative counts^{48,55,56}. Key performance metrics include accuracy (correctly classified data points), precision (correctly

		Depression (n=280, 66.04%)		Anxiety (n=311, 73.35%)		Stress (n=222, 53.35%)	
Variables	Total 424 n (%)	Yes (%)	χ ² (-value)	Yes (%)	χ ² (-value)	Yes (%)	χ ² (-value)
Age							
18–23	196 (46.2)	133 (67.9)	5.274 (0.260)	144 (73.5)	5.881 (0.208)	102 (52.0)	4.413 (0.353)
24–29	195 (46.0)	129 (66.2)		147 (75.4)		104 (53.3)	
30–35	26 (6.1)	13 (50.0)		15 (57.7)		12 (46.2)	
36–41	4 (0.9)	2 (50.0)		2 (50.0)		1 (25.0)	
42–47	3 (0.7)	3 (100.0)		3 (100.0)		3 (100)	
Gender							
Male	290 (68.4)	181 (62.4)	5.373 (0.020) *	207 (71.4)	1.821 (0.177)	141 (48.6)	5.139 (0.023) *
Female	134 (31.6)	99 (73.9)		104 (77.6)		81 (60.4)	
University type							
General	211 (49.8)	151 (71.6)	8.306 (0.040) *	167(79.1)	8.202 (0.042) *	116 (55.0)	2.221 (0.528)
Private	110 (25.9)	70 (63.6)		75 (68.2)		53 (48.2)	
Technical	97 (22.9)	57 (58.8)		66 (68.0)		51 (52.6)	
Medical	6 (1.4)	2 (33.3)		3 (50.0)		2 (33.3)	
Education level							
Graduate	158 (37.3)	106 (67.1)	0.124 (0.725)	117 (74.1)	0.063 (0.801)	86 (54.4)	0.433 (0.510)
Undergraduate	266 (62.7)	174 (65.4)		194 (72.9)		136 (51.1)	
Subject							
Science	212 (50.0)	151 (71.2)	14.028 (0.007) *	170 (80.2)	13.400 (0.009) *	118 (55.7)	4.619 (0.329)
Arts	29 (6.8)	20 (69.0)		22 (75.9)		16 (55.2)	
Commerce	36 (8.5)	28 (77.8)		26 (72.2)		21 (58.3)	
Medical	6 (1.4)	2 (33.3)		3 (50.0)		2 (33.3)	
Engineering	141 (33.3)	79 (56.0)		90 (63.8)		65 (46.1)	
Marital status							
Married	58 (13.7)	32 (55.2)	3.537 (0.060)	37 (36.8)	3.139 (0.076)	24 (41.4)	3.247 (0.072)
Single	366 (86.3)	248 (67.8)		274 (74.9)		198 (54.1)	
Smoking status							
Yes	46 (10.8)	38 (82.6)	6.317 (0.012) *	40 (87.0)	4.887 (0.027) *	35 (76.1)	11.647 (0.001) *
No	378 (89.2)	242 (64.0)		271 (71.7)		187 (49.5)	
No. of respondents' children							
0	403 (95.0)	270 (67.0)	4.701 (0.095)	299 (74.2)	5.916 (0.052)	212 (52.6)	1.421 (0.491)
1	15 (3.5)	6 (40.0)		7 (46.7)		6 (40.0)	
2 or more	6 (1.4)	4 (66.7)		5 (83.3)		4 (66.7)	
Profession							
Government job	32 (7.5)	25 (78.1)	4.295 (0.231)	27 (84.4)	2.566 (0.464)	22 (68.8)	3.851 (0.278)
Private sector	73 (17.2)	43 (58.9)		51 (69.9)		38 (52.1)	
Business	9 (2.1)	7 (77.8)		7 (77.8)		5 (56.6)	
Unemployed	310 (73.1)	205 (66.1)		226 (72.9)		157 (50.6)	
Way of communication							
Direct call	205 (48.3)	137 (66.8)	0.111 (0.739)	153 (74.6)	0.335 (0.563)	111 (54.1)	0.509 (0.476)
Mobile application	219 (51.7)	143 (65.3)		158 (72.1)		111 (50.7)	
Participation in physical activity							
Always	63 (14.9)	39 (61.9)	0.570 (0.752)	46 (73.0)	0.323 (0.851)	28 (44.4)	1.975 (0.372)
Sometimes	300 (70.8)	200 (66.7)		222 (74.0)		160 (53.3)	
Never	61 (14.4)	41 (67.2)		43 (70.5)		34 (55.7)	
Participation in household chores							
Always	122 (28.8)	88 (72.1)	3.276 (0.194)	90 (73.8)	0.040 (0.980)	65 (53.3)	0.142 (0.931)
Sometimes	267 (63.0)	168 (62.9)		195 (73.0)		138 (51.7)	
Never	35 (8.3)	24 (68.6)		26 (74.3)		19 (54.3)	
Memory loss symptoms							
Always	31 (7.3)	26 (83.9)	21.086 (<0.001) *	29 (93.5)	10.195 (0.006) *	22 (71.0)	26.067 (<0.001) *
Sometimes	260 (61.3)	186 (71.5)		194 (74.6)		154 (59.2)	
Never	133 (31.4)	68 (51.1)		88 (66.2)		46 (34.6)	
Continued							

		Depression (n=280, 66.04%)		Anxiety (n=311, 73.35%)		Stress (n=222, 53.35%)	
Variables	Total 424 n (%)	Yes (%)	χ ² (-value)	Yes (%)	χ ² (-value)	Yes (%)	χ ² (-value)
BMI							
Overweight	118 (27.83)	77 (65.3)	1.409 (0.494)	72 (61.0)	1.634 (0.442)	65 (55.1)	0.523 (0.770)
Normal weight	270 (63.68)	176 (65.2)		158 (58.5)		138 (51.1)	
Under weight	36 (8.49)	27 (75.0)		25 (69.4)		19 (52.8)	
Misbehave							
Always	6 (1.4)	6 (100.0)	22.991 (<0.001) *	5 (83.3)	16.526 (<0.001) *	5 (83.3)	37.521 (<0.001) *
Never	193 (45.5)	105 (54.4)		96 (49.7)		70 (36.3)	
Sometimes	225 (53.1)	169 (75.1)		154 (68.4)		147 (65.3)	
Participation in religious activity							
Never	20 (4.7)	18 (90.0)	6.387 (0.041) *	17 (85.0)	1.719 (0.423)	13 (65.0)	1.554 (0.460)
Always	190 (44.8)	134 (62.6)		158 (73.8)		113 (52.8)	
Sometimes	214 (50.5)	128 (67.4)		136 (71.6)		96 (50.5)	
Waking up for bad dreams							
Never	82 (19.3)	38 (46.3)	21.936 (<0.001) *	51 (62.2)	9.466 (0.009) *	32 (39.0)	17.911 (<0.001) *
Always	17 (4.0)	226 (69.5)		244 (75.1)		174 (53.5)	
Sometimes	325 (76.7)	16 (94.1)		16 (94.1)		16 (94.1)	
Take vitamin supplements							
Yes	115 (27.1)	79 (68.7)	0.497 (0.481)	221 (71.5)	1.948 (0.163)	66 (57.4)	1.602 (0.206)
No	309 (72.9)	201 (65.0)		90 (78.3)		156 (50.5)	
Eat nutritious food							
Yes	408 (96.2)	268 (65.7)	0.595 (0.440)	12 (75.0)	0.023 (0.879)	211 (51.7)	1.791 (0.181)
No	16 (3.8)	12 (75.0)		299 (73.3)		11(68.8)	
Google classroom users							
Yes	350 (82.5)	242 (69.1)	8.621 (0.003) *	263 (75.1)	3.301 (0.069)	191 (54.6)	3.937 (0.047) *
No	74 (17.5)	38 (51.4)		48 (64.9)		31 (41.9)	
Anxiety							
Yes	255 (60.1)	261 (83.9)	166.434 (<0.001) *	---		216 (69.5)	136.712 (<0.001) *
No	169 (39.9)	19 (16.8)		---		6 (5.3)	
Stress							
Yes	222 (52.4)	208 (93.7)	158.912 (<0.001) *	216 (69.5)	136.712 (<0.001) *	---	
No	202 (47.6)	72 (35.6)		6 (5.3)		---	
Depression							
Yes	280 (66)	---		261 (83.9)	166.434 (<0.001) *	208 (74.3)	158.91 (<0.001) *
No	144 (34)	---		19 (16.8)		14 (9.7)	

Table 1. Frequency distribution and relationship with depression, anxiety, and stress, and different socio-demographic and behavioral variables among Bangladeshi university students. *Statistically significant at the 0.05 level.

classified positive data points), specificity (correctly classified negative data points), and sensitivity (true positive rate)^{48,55,56}.

Receiver operating characteristic (ROC) Curve

ROC curves are another useful graphical tool for evaluating classifiers using datasets. Fawcett^{50,57} provided a comprehensive introduction to ROC analysis, clarifying common misconceptions. By plotting the true positive rate against the false positive rate, a ROC curve shows the classifier's sensitivity. Higher AUC values indicate better discrimination, with values closer to 1 representing stronger classification performance⁵⁷.

K-fold cross-validation

Cross-validation is a validation method used to evaluate a model's generalization ability on an independent dataset^{48,58,59}. It measures the performance of different predictive functions. In k-fold cross-validation, the training data is randomly divided into k equally sized, mutually exclusive subsamples or folds. The model is trained k times, each time using one-fold for testing and the remaining k-1 folds for training. The results from each iteration are averaged to provide a single estimate of the model's accuracy^{48,58,59}. We applied 10-fold, 20-fold, and 30-fold cross-validation to evaluate classifier performance on this large dataset.

Explainable ML framework

An explainable ML framework, inspired by⁶⁰, was implemented to enhance interpretability of model predictions. In the first stage, predictive models were trained and evaluated to identify the best-performing classifier. In the second stage, post-hoc explainability techniques—SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) were applied to examine model-identified influential features at both global and local levels⁶¹. These explanations describe associations underlying model predictions rather than causal relationships.

Results

Participant characteristics

A total of 424 university students from 38 universities across Bangladesh completed the survey. Nearly half of the respondents were enrolled in general public universities (49.7%), followed by private universities (25.9%), technical universities (22.8%), and medical universities (1.4%). In terms of academic discipline, 50.0% of participants were from science backgrounds, 33.2% from engineering, 8.9% from commerce, 6.8% from arts, and 1.4% from medical studies. Among the participants, 68.4% were male and 31.6% were female. Undergraduate students constituted 62.7% of the sample, while 37.3% were graduate students. Regarding behavioral and demographic characteristics, 10.8% of respondents reported smoking, 23.8% were classified as overweight, 13.6% were married, 7.3% had children, and 26.9% were employed. Detailed descriptive statistics are presented in Table 1.

Distribution and Co-occurrence of depression, anxiety, and stress

Depression, anxiety, and stress were assessed using summated DASS-21 subscale scores, with higher scores indicating greater symptom severity. Based on established cut-offs, 52.35% of respondents were classified as having moderate-to-extremely severe stress, 73.35% as having moderate-to-extremely severe anxiety, and 66.04% as having moderate-to-extremely severe depression (Table 1; Fig. 1a). Substantial overlap among these outcomes was observed. As shown in the Venn diagram (Fig. 1b), 62.02% of students exhibited both anxiety and depression, 49.05% both stress and depression, and 52.36% both anxiety and stress. Additionally, 48.11% met criteria for all three outcomes simultaneously.

Pearson correlation analysis revealed strong positive correlations among continuous scores of depression, anxiety, and stress (Fig. 1c). These correlations may partly reflect overlapping items across DASS-21 subscales rather than entirely independent symptom co-occurrence. Cramer's V analysis of binary classifications indicated significant associations between the outcomes (Fig. 1d). The Sankey diagram (Fig. 1e) provides an additional visualization of these associations. Given the high prevalence of moderate-to-severe symptoms, subsequent machine learning (ML) models should be interpreted as distinguishing relative symptom severity within this high-pathology, self-selected sample, rather than identifying generalizable risk factors.

Associations between mental health outcomes and socio-demographic factors

Chi-square tests were conducted to explore associations between depression, anxiety, and stress and a range of socio-demographic and behavioral variables (Table 1). Depression showed significant associations with gender ($\chi^2 = 5.373$, $p = .02$), university type ($\chi^2 = 8.306$, $p = .01$), academic discipline ($\chi^2 = 14.028$, $p < .001$), smoking status ($\chi^2 = 6.317$, $p = .02$), memory loss symptoms ($\chi^2 = 21.086$, $p < .001$), participation in religious activities ($\chi^2 = 6.387$, $p = .01$), misbehavior ($\chi^2 = 22.991$, $p < .001$), waking up from bad dreams ($\chi^2 = 21.936$, $p < .001$), and Google Classroom usage ($\chi^2 = 8.621$, $p = .003$). Higher proportions of depressive symptoms were observed among female students (60.4%), smokers (76.1%), students enrolled in public universities (71.6%), individuals reporting frequent bad dreams (94.1%), government employees (78.1%), and those not participating in religious activities (90.0%).

Anxiety was significantly associated with university type ($\chi^2 = 8.202$, $p = .01$), academic discipline ($\chi^2 = 13.400$, $p < .001$), smoking status ($\chi^2 = 4.887$, $p = .03$), memory loss symptoms ($\chi^2 = 10.195$, $p = .002$), misbehavior ($\chi^2 = 16.526$, $p < .001$), and waking up from bad dreams ($\chi^2 = 9.466$, $p = .002$). Higher anxiety prevalence was observed among female students (77.6%), those aged 24–29 years (75.4%), students attending general public universities (79.1%), smokers (87.0%), individuals reporting sleep disturbances (94.1%), and government employees (84.4%).

Stress was significantly associated with gender ($\chi^2 = 5.139$, $p = .02$), memory loss symptoms ($\chi^2 = 26.067$, $p < .001$), misbehavior ($\chi^2 = 37.521$, $p < .001$), waking up from bad dreams ($\chi^2 = 17.911$, $p < .001$), and Google Classroom usage ($\chi^2 = 3.937$, $p = .05$). Elevated stress levels were particularly prevalent among students reporting frequent nightmares (94.1%), smoking behavior (76.1%), and misbehavior (83.3%). Elevated stress was also common among government employees (68.8%), those not participating in religious activities (65.0%), and students aged 24–29 (53.3%). Female students (60.4%) also reported significantly higher stress levels compared to male students (48.6%). These associations reflect statistical relationships observed within the sample and do not imply causality.

Feature selection

Feature selection was performed using both traditional statistical testing and machine-learning-based methods. Based on chi-square analyses, nine variables were significantly associated with depression, six with anxiety, and six with stress (Table 1). To complement these findings and reduce model dependency on uninformative predictors, the Boruta algorithm was applied to identify robust features for each mental health outcome (Fig. 2).

Boruta analysis identified waking up from bad dreams, memory loss symptoms, participation in religious activities, misbehavior, smoking status, and academic discipline as important predictors of depression, with Google Classroom usage classified as a tentative feature. For anxiety prediction, academic discipline, severe internet addiction (IAT), misbehavior, waking up from bad dreams, and memory loss symptoms were identified as important features, while smoking status and number of children were tentatively selected. For stress prediction, misbehavior, memory loss symptoms, smoking status, and waking up from bad dreams emerged as

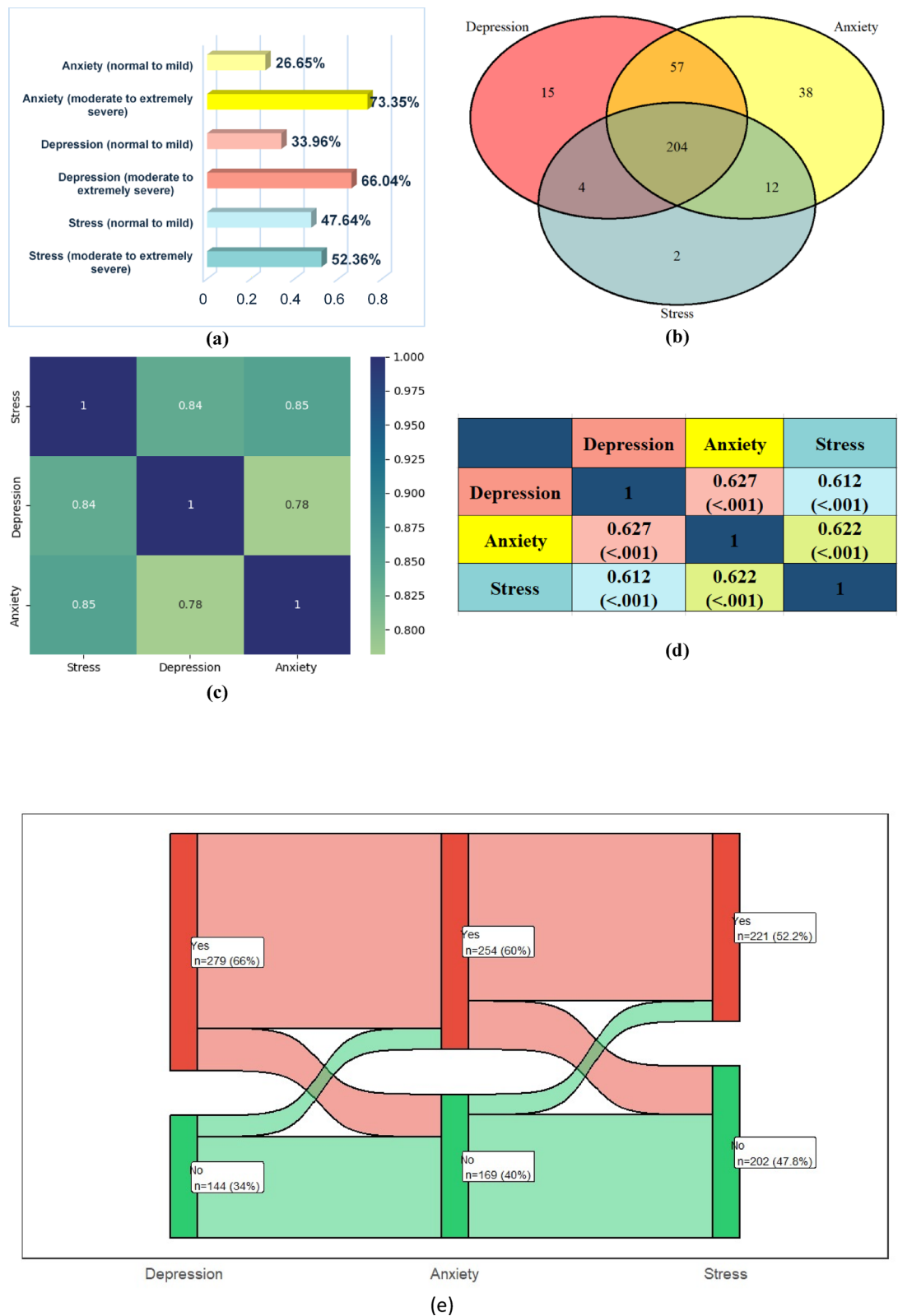


Fig. 1. Bar diagram demonstrating the prevalence of stress, anxiety, and depression in (a); Venn diagram illustrating the overlapping of depression, anxiety, and stress in (b); Correlation heatmap between depression, anxiety, and stress based on Pearson correlation coefficient in (c); association of binary responses of depression, anxiety, and stress using Cramer's V in (d); and Sankey diagrams illustrating the relationships between depression, anxiety, and stress in (e) among the surveyed university students.

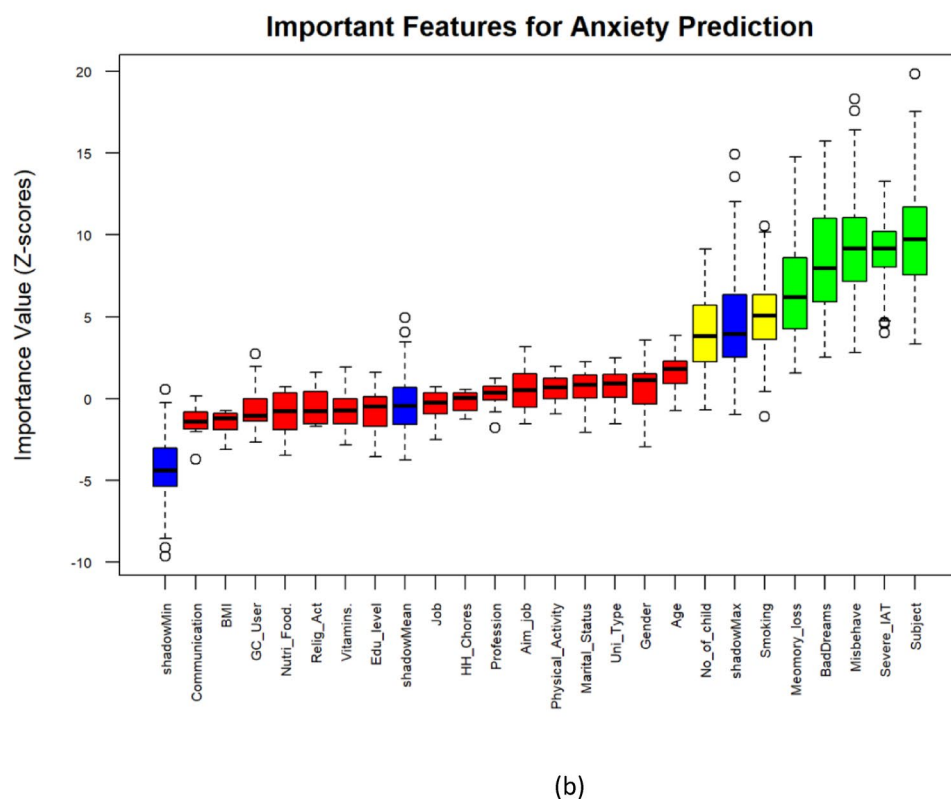
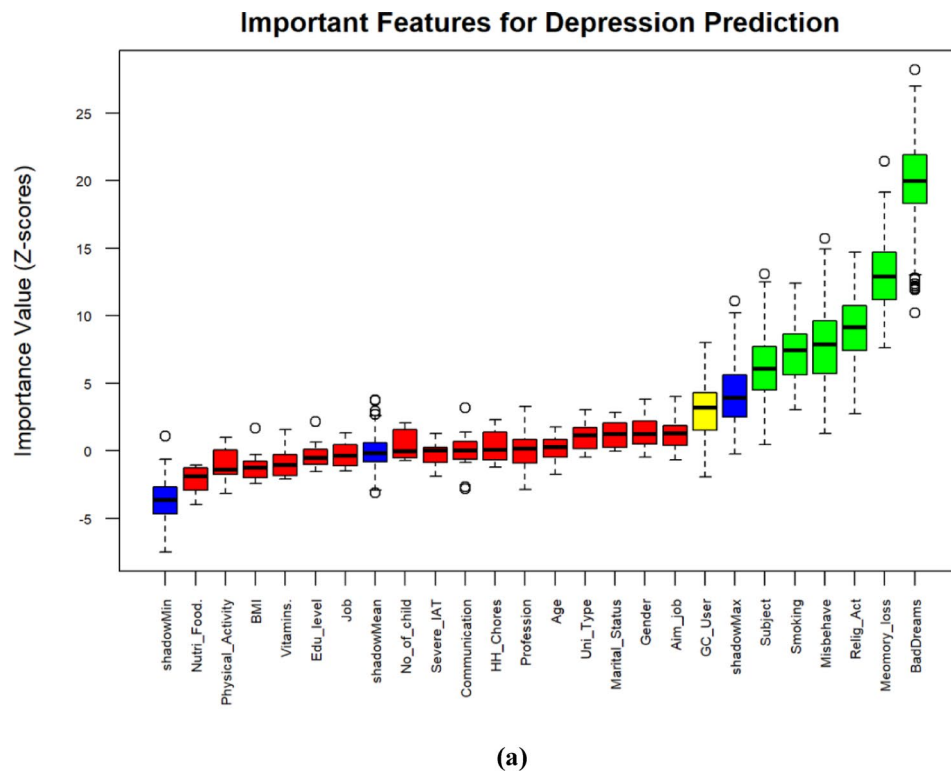


Fig. 2. Selected features based on the Boruta algorithm to predict depression in (a), anxiety in (b), and stress in (c) among university students.

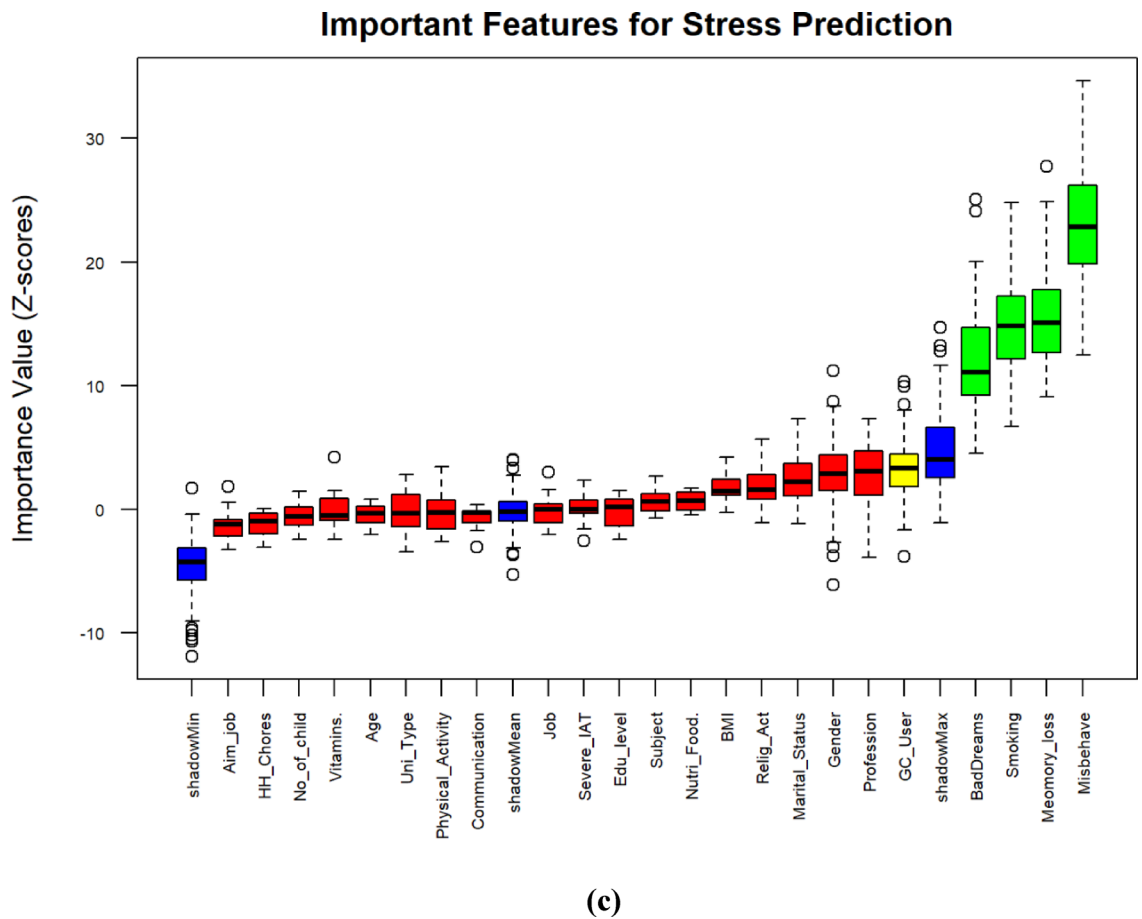


Fig. 2. (continued)

the most influential predictors, with Google Classroom usage again tentatively selected. Based on the combined evidence from chi-square testing and Boruta selection, final feature sets were defined separately for depression, anxiety, and stress prediction models.

Performance of machine learning (ML) models

Eight ML models, including DT, RF, LR, SVM, AdaBoost, XGBoost, LightGBM, and ANNs, were evaluated for predicting depression, anxiety, and stress. Hyperparameters were optimized using grid search (Table 2). The dataset was repeatedly split into training (70%) and testing (30%) sets across 1,000 permutations to ensure robustness. Model performance was assessed using accuracy, sensitivity, specificity, precision, F1-score, cross-validation accuracy, and AUC (Table 3; Fig. 3).

For depression prediction, the SVM demonstrated a relatively consistent balance of classification performance, achieving a sensitivity of 0.8551, precision of 0.7320, overall accuracy of 0.6959, and an AUC of 0.6976. AdaBoost achieved the highest sensitivity (0.9696) and F1-score (0.7942), indicating excellent capability in identifying individuals with depressive symptoms; however, this gain was accompanied by reduced specificity. XGBoost attained the highest AUC (0.7019), reflecting superior overall discriminative ability across classification thresholds. Importantly, SVM outperformed all other models in cross-validation, yielding the highest accuracy in both 10-fold (0.7099) and 30-fold (0.7080) validations. The stability of SVM performance across multiple validation schemes indicates relatively consistent performance and limited evidence of overfitting within the study sample. Considering its relatively balanced performance across metrics and more consistent cross-validation results compared with other models, SVM was selected as the preferred model for depression prediction.

In the context of anxiety prediction, XGBoost achieved the highest AUC (0.6852) among the evaluated models; however, the magnitude of discrimination remained poor to fair. AdaBoost recorded the highest accuracy (0.7332), sensitivity (0.9999), and F1-score (0.8456); however, its specificity was extremely low (0.00002), indicating a near-complete inability to correctly identify non-anxious individuals. Such a performance profile, while favorable for case detection, substantially limits practical utility due to an excessive false-positive rate. By contrast, XGBoost provided a more balanced classification performance, with an accuracy of 0.7151, sensitivity of 0.8856, precision of 0.7648, F1-score of 0.8195, and markedly higher specificity (0.2521). This balance between sensitivity and specificity is theoretically important for mental health screening applications, although the absolute performance levels observed here remain insufficient for practical deployment. Accordingly, XGBoost

Model	Hyperparameter	Search Range	Tuned Value
Logistic Regression	C	[0.01, 10]	1.01
	max_iter	[10, 50]	50
	penalty	['l1', 'l2', 'elasticnet', 'none']	'l2'
	solver	['newton-cg', 'lbfgs', 'liblinear', 'sag', 'saga']	'saga'
XGBoost	learning_rate	[0.01, 0.3]	0.1
	max_depth	[3, 10]	4
	n_estimators	[50, 1000]	100
	subsample	[0.5, 1.0]	1.0
	colsample_bytree	[0.5, 1.0]	1.0
LightGBM	learning_rate	[0.01, 0.3]	0.1
	num_leaves	[20, 300]	31
	max_depth	[-1, 15]	-1
	n_estimators	[50, 1000]	100
	subsample	[0.5, 1.0]	1.0
AdaBoost	n_estimators	[50, 500]	100
	learning_rate	[0.01, 2.0]	0.1
	algorithm	['SAMME', 'SAMME.R']	'SAMME.R'
Decision tree	max_depth	[None, 20]	5
	min_samples_split	[2, 20]	2
	min_samples_leaf	[1, 20]	1
	criterion	['gini', 'entropy']	'gini'
Random forest	n_estimators	[50, 500]	100
	max_depth	[None, 20]	5
	min_samples_split	[2, 20]	2
	min_samples_leaf	[1, 20]	1
	max_features	['auto', 'sqrt', 'log2', None]	'auto'
SVM	C	[0.01, 100]	0.1
	max_iter	[10, 50]	50
	kernel	['linear', 'poly', 'rbf', 'sigmoid']	'poly'
	degree	[2, 5]	3
	gamma	['scale', 'auto'] or float	'scale'
ANN (MLP)	hidden_layer_sizes	Various architectures	(10, 5)
	activation	['relu', 'tanh', 'logistic']	'relu'
	solver	['adam', 'lbfgs', 'sgd']	'adam'
	alpha	[0.0001, 0.1]	0.01
	learning_rate	['constant', 'invscaling', 'adaptive']	'constant'
	max_iter	[100, 2000]	1000

Table 2. Hyperparameter tuning using the Grid search for the DT, RF, LR, SVM, AdaBoost, XGBoost, LightGBM, and ANNs models to predict MHPs among university students in Bangladesh.

was retained for anxiety prediction due to its comparatively better balance across metrics; however, overall predictive performance for anxiety remained modest, and none of the evaluated models demonstrated adequate discriminative ability.

Stress prediction posed greater difficulty than depression and anxiety, as evidenced by generally lower accuracy across all evaluated models, with no model exceeding an accuracy of 0.664 in repeated testing (Table 3). Despite this challenge, SVM consistently demonstrated superior performance relative to other models. Specifically, SVM achieved the highest cross-validated accuracies in 10-fold (0.6765), 20-fold (0.6816), and 30-fold (0.6785) validations, alongside the highest F1-score (0.6883) and AUC (0.7238). The comparatively higher AUC indicates a modest improvement in SVM's ability to discriminate between stressed and non-stressed individuals across decision thresholds, while the consistent cross-validation performance suggests reliable generalization within the observed sample. Based on these results, SVM was selected as the relatively better-performing model for stress prediction within the evaluated set, despite overall modest classification accuracy.

Overall, Table 3; Fig. 3 demonstrate that SVM provides relatively more consistent and balanced performance for predicting depression and stress compared with other evaluated models, whereas XGBoost shows comparatively higher but still limited discriminative capability for anxiety prediction. Although performance levels varied across outcomes, the selected models achieved consistent and interpretable predictive accuracy across repeated permutations and validation schemes. These findings support the use of explainable machine

	Metric	DT	RF	LR	SVM	XGBoost	AdaBoost	LightGBM	ANNs
Depression	Accuracy	0.6581 (0.0380)	0.6778 (0.0352)	0.6945 (0.0354)	0.6959 (0.0356)	0.6903 (0.0357)	0.6692 (0.0373)	0.6604 (0.0348)	0.6713 (0.0359)
	Sensitivity	0.7215 (0.0552)	0.7889 (0.0491)	0.8914 (0.0435)	0.8551 (0.0446)	0.8297 (0.0476)	0.9696 (0.0357)	0.8055 (0.0525)	0.7999 (0.0486)
	Specificity	0.5376 (0.0771)	0.4656 (0.0795)	0.3156 (0.0727)	0.3904 (0.0798)	0.4228 (0.0792)	0.0876 (0.0692)	0.3822 (0.0769)	0.4240 (0.0796)
	Precision	0.7525 (0.0449)	0.7420 (0.0439)	0.7171 (0.0418)	0.7320 (0.0437)	0.7369 (0.0431)	0.6744 (0.0399)	0.7175 (0.0425)	0.7303 (0.0430)
	F1-score	0.7348 (0.0359)	0.7631 (0.0300)	0.7934 (0.0274)	0.7873 (0.0284)	0.7790 (0.0295)	0.7942 (0.0265)	0.7572 (0.0300)	0.7619 (0.0304)
	AUC	0.6378 (0.0451)	0.6929 (0.0413)	0.6914 (0.0434)	0.6976 (0.0414)	0.7019 (0.0428)	0.6807 (0.0444)	0.6734 (0.0416)	0.6662 (0.0442)
	10-Fold	0.6654 (0.0744)	0.6793 (0.0647)	0.6961 (0.0690)	0.7099 (0.0556)	0.6583 (0.0706)	0.6744 (0.0535)	0.6700 (0.0721)	0.7051 (0.0412)
	20-Fold	0.6615 (0.0999)	0.6708 (0.1037)	0.7061 (0.1107)	0.7033 (0.1066)	0.6709 (0.1039)	0.6747 (0.0949)	0.6614 (0.1195)	0.6703 (0.0922)
	30-Fold	0.6566 (0.1207)	0.6847 (0.1228)	0.7060 (0.1260)	0.7080 (0.1192)	0.6734 (0.1428)	0.6795 (0.1022)	0.6358 (0.1438)	0.6773 (0.1083)
Anxiety	Accuracy	0.6769 (0.0387)	0.7025 (0.0348)	0.7248 (0.335)	0.7193 (0.0337)	0.7151 (0.0336)	0.7332 (0.0335)	0.7225 (0.0335)	0.7076 (0.0348)
	Sensitivity	0.7909 (0.0492)	0.8613 (0.0418)	0.9540 (0.0312)	0.9514 (0.0446)	0.8856 (0.0403)	0.9999 (0.0017)	0.8961 (0.0393)	0.8792 (0.0454)
	Specificity	0.3684 (0.0878)	0.2710 (0.0816)	0.0986 (0.0556)	0.0869 (0.0754)	0.2521 (0.0825)	0.00002 (0.0008)	0.2506 (0.0809)	0.2407 (0.0815)
	Precision	0.7747 (0.0416)	0.7644 (0.0394)	0.7441 (0.0364)	0.7412 (0.0377)	0.7648 (0.0397)	0.73325 (0.0335)	0.7666 (0.0387)	0.7609 (0.0382)
	F1-score	0.7813 (0.0308)	0.8087 (0.0259)	0.8352 (0.0229)	0.8318 (0.0239)	0.8195 (0.0245)	0.8456 (0.0225)	0.8251 (0.0242)	0.8144 (0.0259)
	AUC	0.6227 (0.0503)	0.6471 (0.0469)	0.6758 (0.0445)	0.6785 (0.0472)	0.6852 (0.0457)	0.6188 (0.0505)	0.6527 (0.0476)	0.6561 (0.0450)
	10-Fold	0.6748 (0.0702)	0.7033 (0.0802)	0.7169 (0.0534)	0.7193 (0.0687)	0.7289 (0.0665)	0.7335 (0.0707)	0.7383 (0.0665)	0.7242 (0.0827)
	20-Fold	0.6748 (0.0870)	0.7173 (0.0888)	0.7147 (0.0909)	0.7218 (0.0891)	0.7217 (0.0924)	0.7336 (0.0966)	0.7265 (0.0949)	0.7336 (0.0966)
	30-Fold	0.6802 (0.1277)	0.6985 (0.1280)	0.7198 (0.0885)	0.7269 (0.1016)	0.7174 (0.0992)	0.7341 (0.1062)	0.7292 (0.1131)	0.7246 (0.1044s)
Stress	Accuracy	0.6493 (0.0365)	0.6469 (0.0357)	0.6453 (0.0375)	0.6596 (0.0355)	0.6582 (0.0342)	0.6590 (0.0478)	0.6638 (0.0354)	0.6519 (0.0365)
	Sensitivity	0.6091 (0.0682)	0.6433 (0.0709)	0.7298 (0.0598)	0.7037 (0.0631)	0.6352 (0.0619)	0.6974 (0.0730)	0.6447 (0.0634)	0.6434 (0.0693)
	Specificity	0.6969 (0.0831)	0.6545 (0.0870)	0.5550 (0.0716)	0.6138 (0.0738)	0.6871 (0.0749)	0.6235 (0.1220)	0.6885 (0.0758)	0.6649 (0.0867)
	Precision	0.6921 (0.0644)	0.6738 (0.0616)	0.6449 (0.0523)	0.6691 (0.0544)	0.6932 (0.0603)	0.6768 (0.0764)	0.6974 (0.0609)	0.6822 (0.0625)
	F1-score	0.6438 (0.0431)	0.6547 (0.0411)	0.6822 (0.0387)	0.6883 (0.0381)	0.6595 (0.0388)	0.6812 (0.0415)	0.6665 (0.0398)	0.6581 (0.0405)
	AUC	0.6857 (0.0394)	0.7013 (0.0379)	0.6947 (0.0401)	0.7238 (0.0383)	0.7151 (0.0368)	0.7043 (0.0390)	0.7159 (0.0381)	0.7021 (0.0385)
	10-Fold	0.6720 (0.0434)	0.6555 (0.0509)	0.6368 (0.0431)	0.6765 (0.0845)	0.6578 (0.0654)	0.6504 (0.0623)	0.6625 (0.0657)	0.6648 (0.0642)
	20-Fold	0.6629 (0.0694)	0.6511 (0.0879)	0.6490 (0.0612)	0.6816 (0.1091)	0.6558 (0.0846)	0.6815 (0.0828)	0.6561 (0.0976)	0.5802 (0.0978)
	30-Fold	0.6583 (0.1261)	0.6536 (0.1228)	0.6539 (0.1305)	0.6785 (0.1612)	0.6488 (0.1305)	0.6811 (0.1281)	0.6561 (0.1281)	0.6155 (0.1352)

Table 3. Accuracy, sensitivity, specificity, precision, F1-score, and AUC of different ML models to predict depression, anxiety, and stress based on 1000 permutations.

learning frameworks as pattern-recognition and risk-stratification tools within the study sample, rather than as population-level diagnostic models, aligning with the study's methodological scope.

Explainable ML analysis of model predictions for stress, anxiety, and depression

An explainable machine learning (XAI) framework was applied to examine features most strongly associated with model predictions of depression, anxiety, and stress among university students. Support vector machine (SVM) models were used for depression and stress prediction, while an extreme gradient boosting (XGBoost) model was used for anxiety prediction. Model interpretability was explored using both global and local explanation

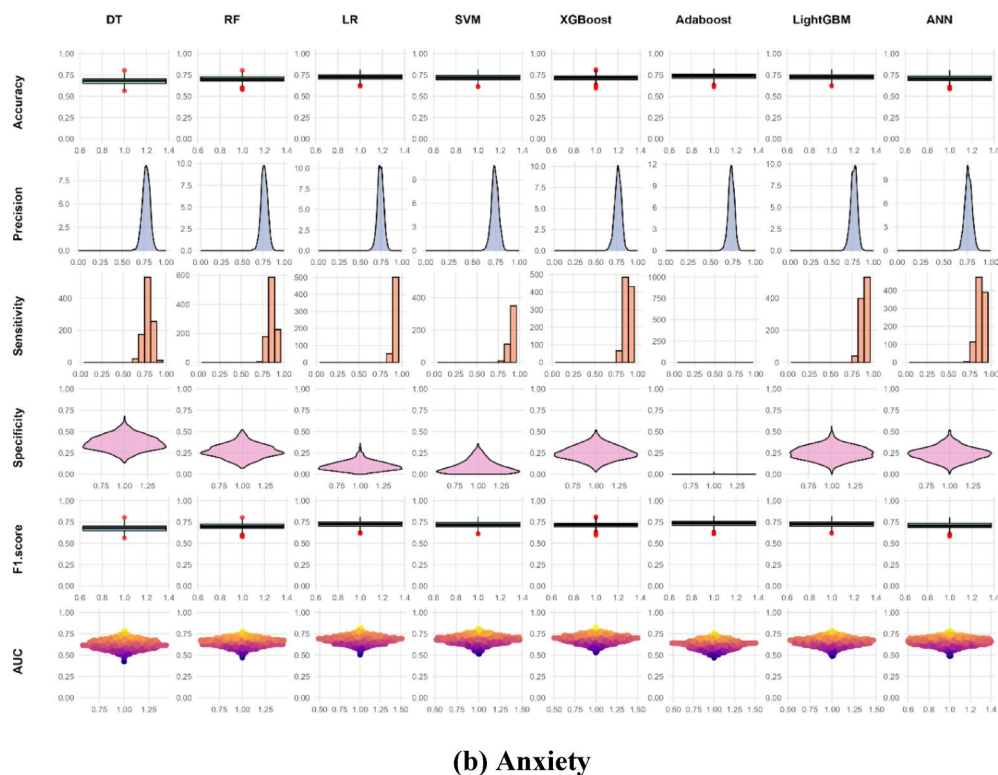
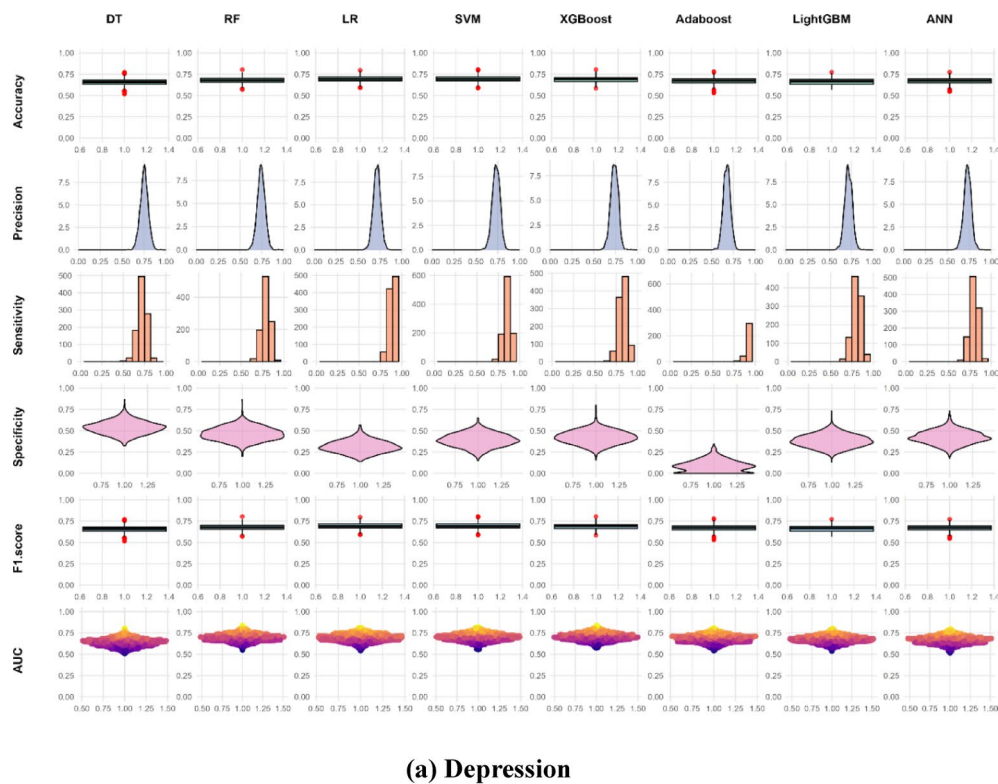
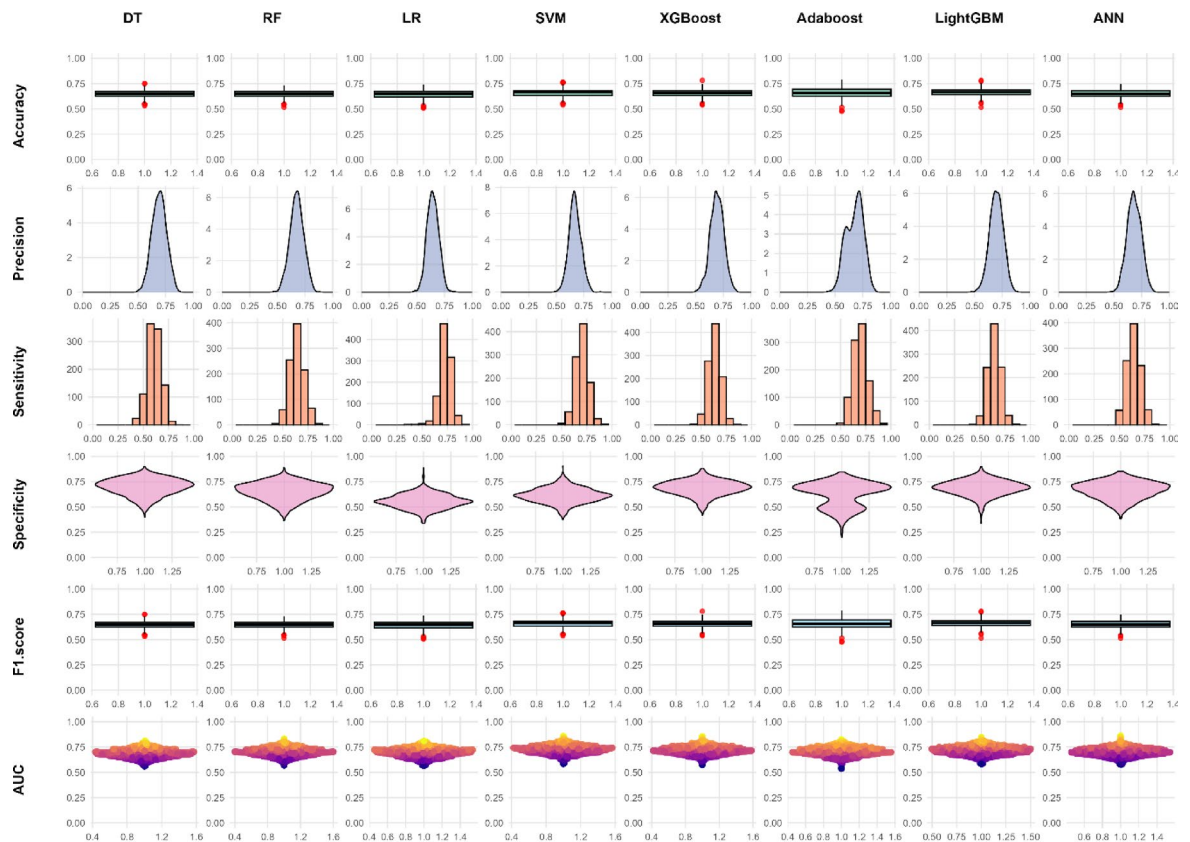


Fig. 3. Performance comparison of eight ML models (DT, RF, LR, SVM, XGBoost, Adaboost, LightGBM, and ANN) across six different evaluation metrics for the prediction of (a) depression, (b) anxiety, and (c) stress. Each cell displays the distribution of metric values for the respective model based on 1000 permutations run using different visualization types (boxplot, density plot, histogram, violin plot, beeswarm).



(c) Stress

Fig. 3. (continued)

techniques, including SHAP and LIME to enhance transparency of the prediction mechanisms without implying causal relationships.

SHAP-based global and local interpretations

Figure 4 presents SHAP-based explanations for depression, anxiety, and stress predictions. For each outcome, a global feature importance bar plot displays the mean absolute SHAP values, which quantify the relative contribution of each feature to the model's prediction output. These values are model-specific and comparative and should not be interpreted as effect sizes, causal impacts, or indicators of clinical importance. For depression prediction (Fig. 4a), the highest mean absolute SHAP values were observed for misbehavior (0.576), waking up from bad dreams (0.543), academic discipline (0.535), memory loss symptoms (0.531), and participation in religious activities (0.403). Higher values of misbehavior, sleep disturbance, and cognitive complaints were generally associated with higher predicted probabilities of depression within the model, although feature effects varied across individuals, indicating heterogeneity in prediction patterns.

For anxiety prediction (Fig. 4b), academic discipline exhibited the largest mean absolute SHAP value (0.583), followed by misbehavior (0.417), university type (0.306), waking up from bad dreams (0.294), and memory loss symptoms (0.283). These features collectively shaped anxiety prediction outputs, with SHAP values indicating both linear and nonlinear contributions depending on feature levels. For stress prediction (Fig. 4c), the largest SHAP contributions were observed for misbehavior (0.622), memory loss symptoms (0.456), and waking up from bad dreams (0.314). Higher values of these features were consistently associated with increased stress prediction scores, while other predictors exhibited smaller or context-dependent contributions.

The SHAP summary plots further illustrate that features related to sleep disruption, cognitive complaints, and behavioral factors play prominent roles in shaping model predictions across all three outcomes, likely reflecting overlapping symptom dimensions rather than independent predictors. Given the cross-sectional design and the symptom-based nature of several predictors, these patterns likely reflect overlapping symptom dimensions captured by the models rather than independent antecedent risk factors.

LIME-based local explanations

Figure 5 illustrates LIME-based local explanations for individual predictions of depression, anxiety, and stress. Each panel depicts how specific feature values contribute positively or negatively to a given classification outcome for representative cases. Across all three outcomes, features such as waking up from bad dreams, memory loss

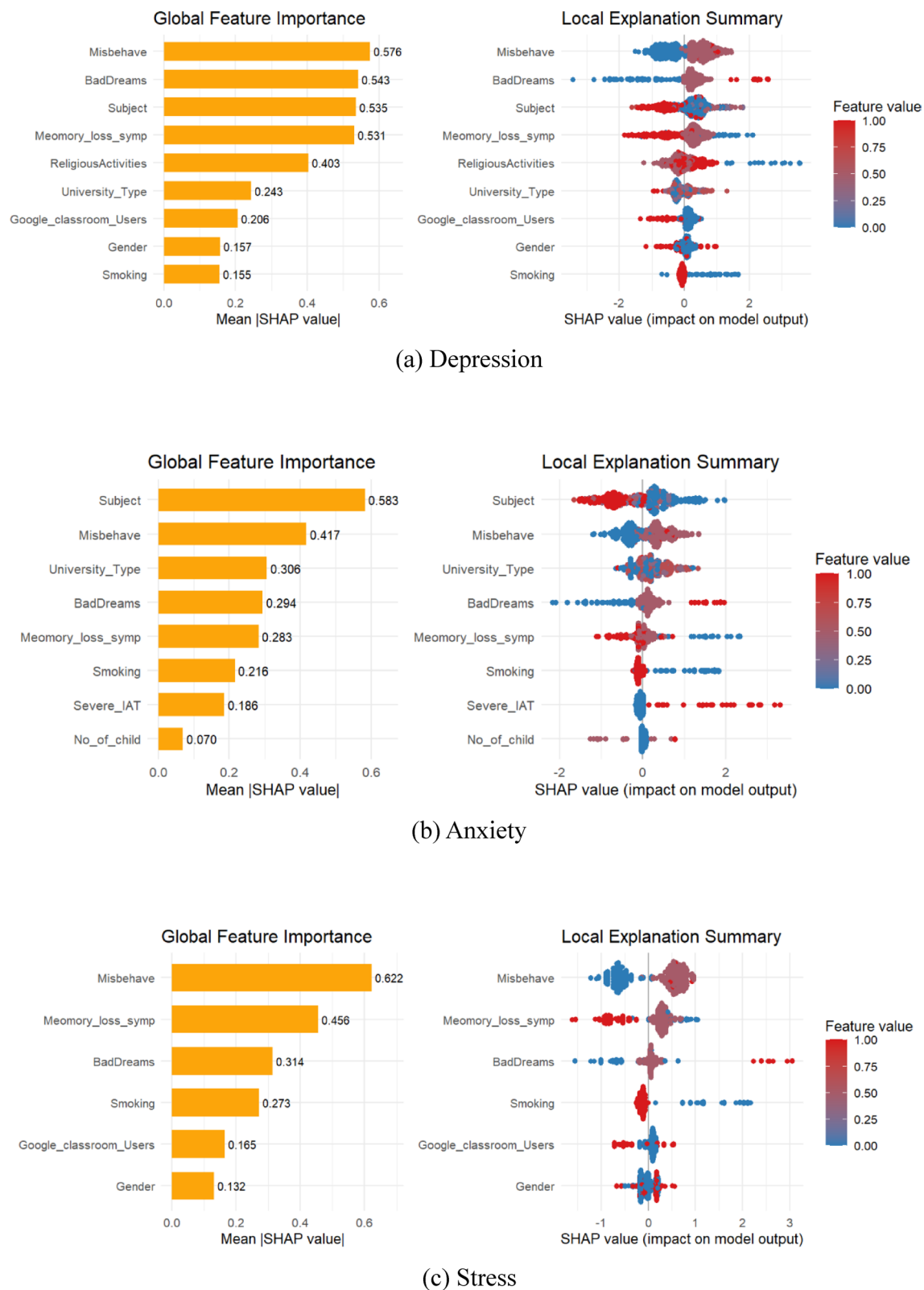
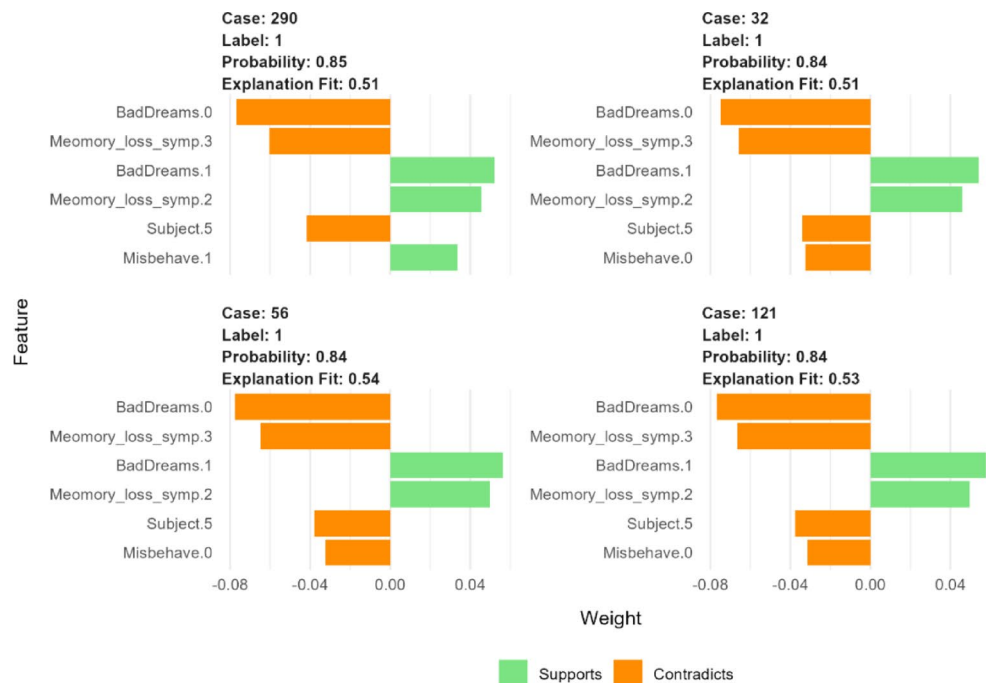
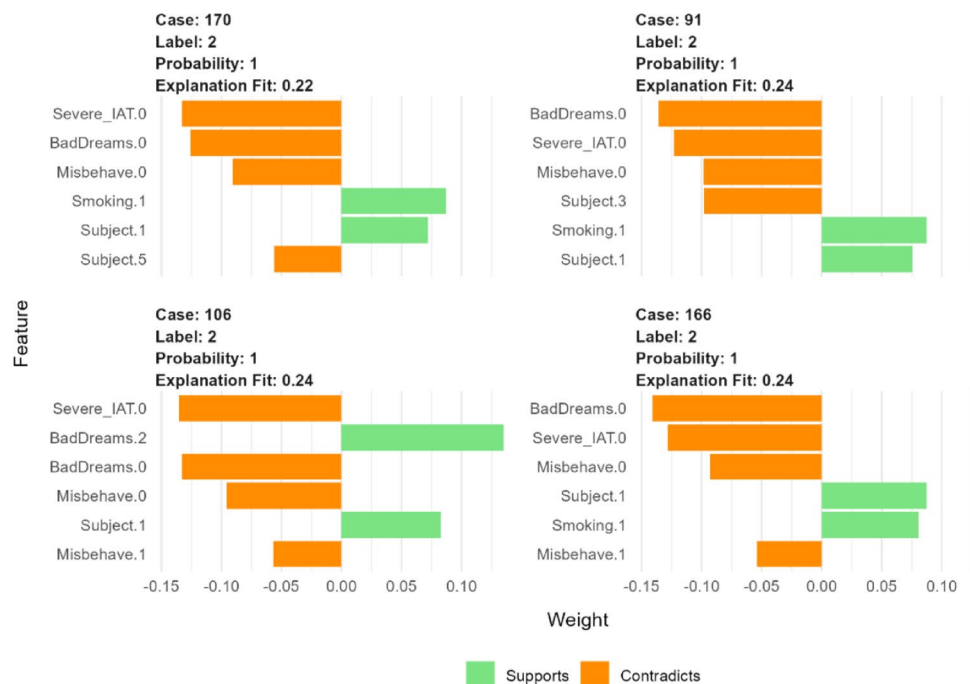


Fig. 4. SHAP-Based Feature Importance and Local Explanation Summaries for Predicting Depression in (a), Anxiety in (b), and Stress in (c) among University Students in Bangladesh.



(a) Depression



(b) Anxiety

Fig. 5. LIME-based local explanations for model predictions of (a) depression, (b) anxiety, and (c) stress. The models were trained using support vector machines (SVM) for depression and stress, and XGBoost for anxiety, following feature sets specified by domain expertise. Each subpanel illustrates how specific feature values support (green) or contradict (orange) the predicted label for individual cases.

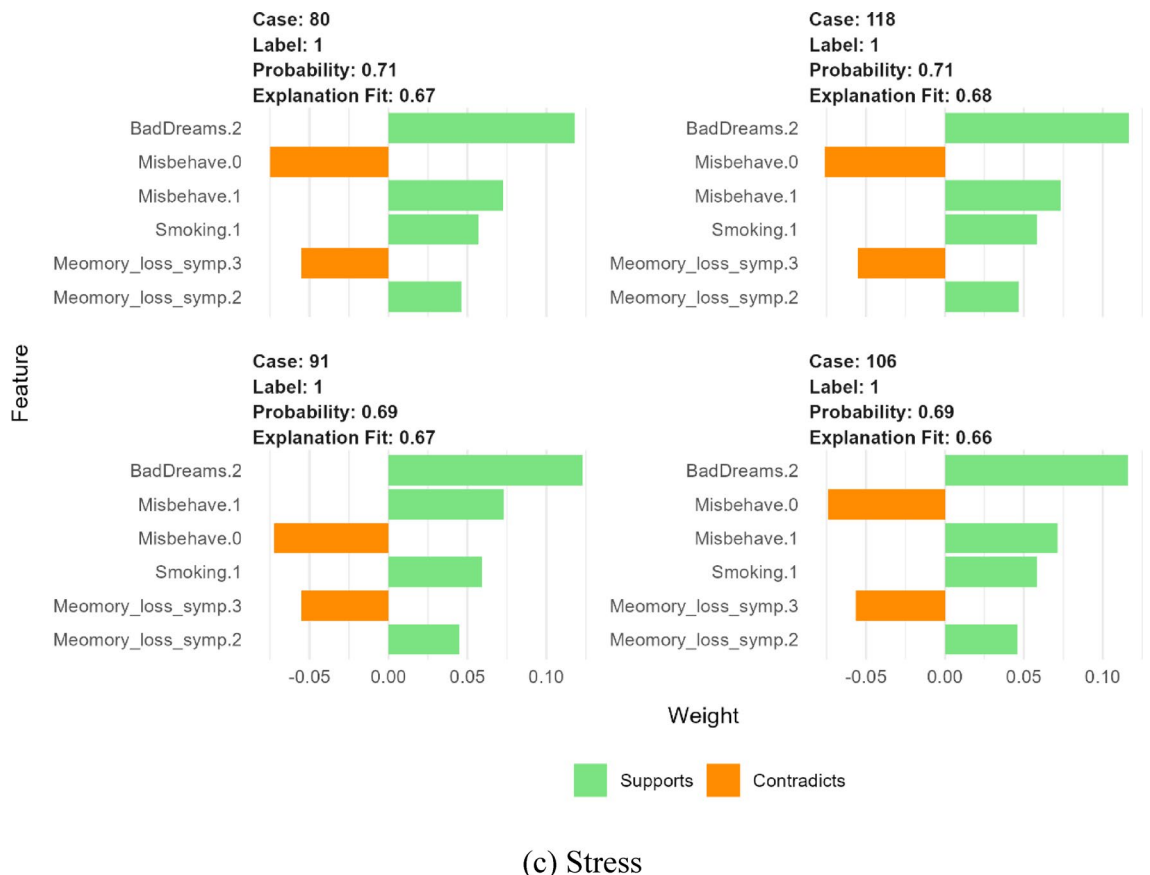


Fig. 5. (continued)

symptoms, and misbehavior frequently contributed to higher predicted risk classifications, whereas other features exerted variable effects depending on individual profiles. For depression predictions, sleep-related symptoms and cognitive complaints showed the strongest local contributions. Anxiety predictions were additionally influenced by smoking status, internet addiction severity, and academic discipline, reflecting the combined influence of behavioral, lifestyle, and academic context within the model. Stress predictions were primarily shaped by sleep disturbance, smoking behavior, and behavioral dysregulation, underscoring the importance of psychosomatic and behavioral features in stress-related prediction patterns.

Sankey diagram visualization of feature–outcome associations

Figure 6 presents Sankey diagrams visualizing the relative strength of model-derived associations between key features and predicted mental health outcomes. Across depression, anxiety, and stress, waking up from bad dreams, memory loss symptoms, and misbehavior consistently appeared as prominent feature–outcome linkages. While each outcome exhibited a distinct configuration of influential features, the repeated prominence of sleep-related and cognitive symptoms suggests substantial overlap in the symptom dimensions captured by the models. Taken together, the explainable ML analyses demonstrate that the predictive models primarily leverage features reflecting overlapping symptomatology and behavioral correlates rather than independent antecedent risk factors. These findings support the use of explainable machine learning frameworks primarily as pattern-recognition tools within this high-symptom study sample, rather than as accurate screening or population-level risk prediction models.

Discussion

Mental health is a multidimensional construct encompassing emotional, psychological, and social well-being, influencing cognition, behavior, interpersonal functioning, and stress coping across the life course. University students represent a particularly vulnerable population, given academic pressure, transitional life stages, and socio-economic uncertainty. Previous studies have consistently documented elevated levels of depression, anxiety, and stress among university students globally^{6–8,10–12}, and growing evidence indicates that mental health problems (MHPs) among Bangladeshi students are of increasing public health concern, with serious adverse associations reported in the literature, including elevated suicide risk and links to cardiovascular outcomes such as heart disease and stroke^{19,20,62,63}. In this context, machine learning (ML) approaches offer potential value for pattern recognition and risk stratification within complex, high-dimensional mental health data^{21–25}. However, applications of explainable ML for student mental health in Bangladesh remain limited. This study adds to the

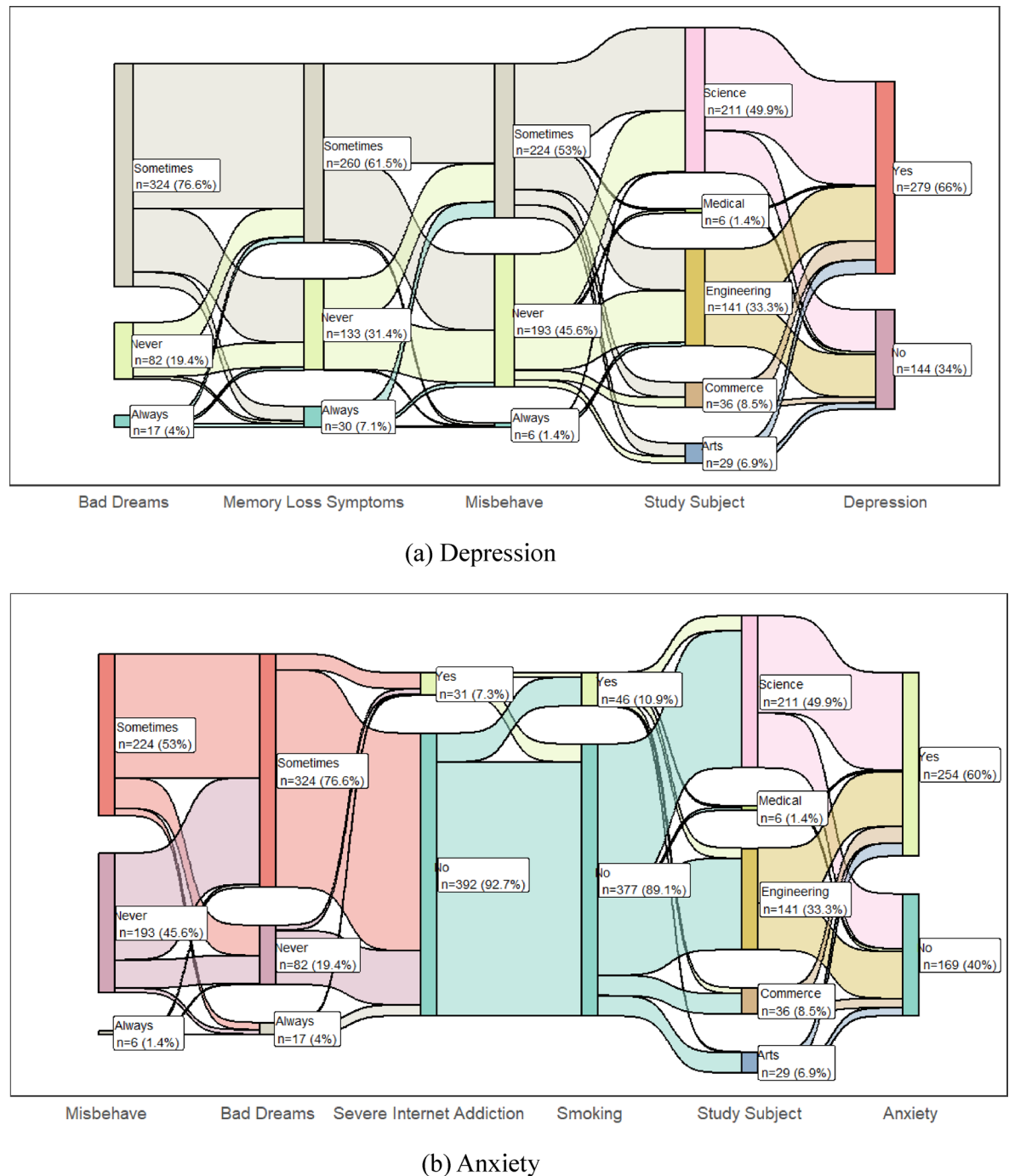
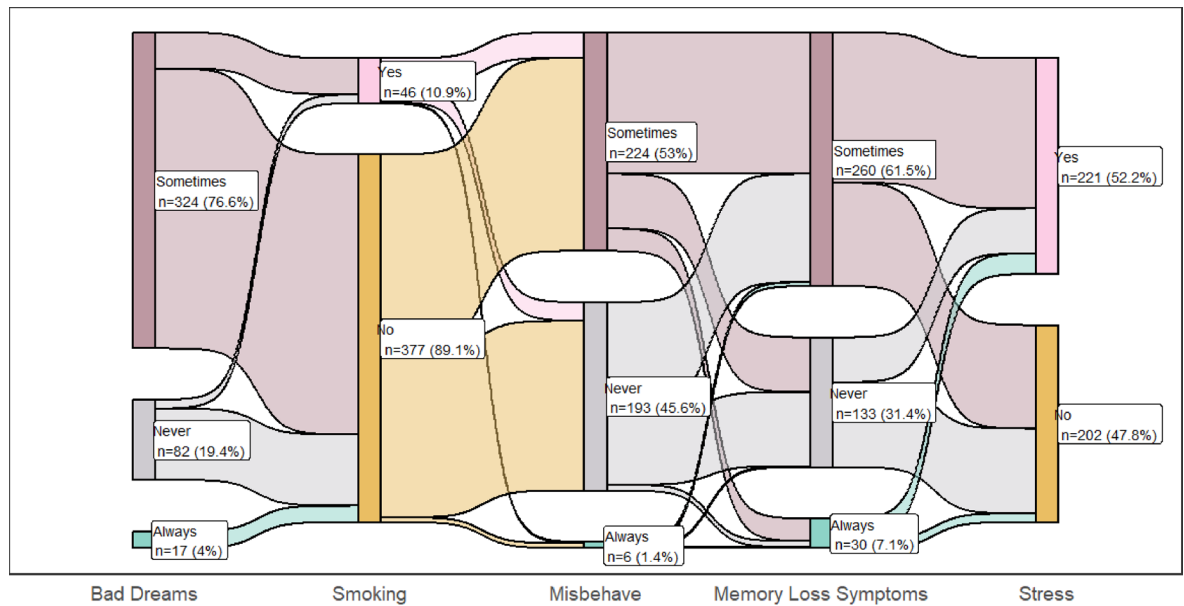


Fig. 6. Sankey diagrams illustrating the relationships between key predictors and mental health outcomes: **(a)** Depression, **(b)** Anxiety, and **(c)** Stress.

existing literature by examining patterns of depression, anxiety, and stress within a high-symptom student sample using an explainable ML framework, while explicitly acknowledging methodological and interpretive constraints.

The present study observed a high prevalence of moderate to extremely severe depression (66.04%), anxiety (73.35%), and stress (52.35%) among surveyed university students. These estimates are higher than those reported in several prior studies among university students using the DASS-21 scale^{6–9,11,12,14}; however, direct comparisons are limited by differences in sampling methods and study context. Importantly, these figures reflect symptom severity within a self-selected, cross-sectional online sample rather than population-level prevalence. Cross-country prevalence comparisons were not conducted, as our self-selected sample may not represent population-level estimates and could be misleading if directly compared to representative surveys. The timing of data collection coincided with the July 2024 student movement in Bangladesh, a period marked by political



(c) Stress

Fig. 6. (continued)

unrest, violence, academic disruption, and heightened uncertainty^{64,65}. Exposure to such stressors may be associated with elevated psychological distress, and therefore the observed prevalence should be interpreted as context-specific rather than representative of typical student mental health conditions. Strong positive correlations and substantial overlap were observed among depression, anxiety, and stress, with nearly half of respondents meeting criteria for all three conditions simultaneously. This degree of co-occurrence aligns with prior literature documenting shared symptom dimensions and high comorbidity among common mental health problems^{7,8,10}. These findings reinforce the conceptual interrelatedness of these constructs and underscore the importance of considering them jointly rather than as isolated outcomes. The predictive analysis for continuous scores of depression, anxiety, and stress, presented in Supplementary Table S1, reveals that even when retaining the full information content of the DASS-21 scores, predictive performance remains consistently low across all models ($R^2 < 0.25$ for all outcomes). This finding underscores the inherent limitations of predicting mental health outcomes from cross-sectional, self-reported behavioral and demographic features. The primary contribution of this study is therefore not predictive accuracy, but rather the transparent application of an explainable ML framework to student mental health data in Bangladesh, providing a methodological template for future research, identifying patterns of feature associations, and establishing benchmark performance metrics.

Several socio-demographic and behavioral variables showed statistically significant associations with depression, anxiety, and stress within the sample, including gender, university type, academic discipline, smoking status, sleep disturbance, memory-related complaints, and self-reported behavioral difficulties. These associations broadly align with patterns reported in earlier studies reporting higher symptom burden among female students^{6,7,11,66}, smokers^{9,67}, and individuals reporting sleep or cognitive disturbances^{68–72}. However, given the cross-sectional design, these relationships should be interpreted strictly as associative rather than causal. Many identified variables—such as waking up from bad dreams, memory loss symptoms, and misbehavior—are likely manifestations or correlates of psychological distress rather than independent antecedent risk factors.

The explainable ML analyses using SHAP, LIME, and Sankey diagrams illustrated how models weighted different features when generating predictions. Across depression, anxiety, and stress, features related to sleep disturbance, cognitive complaints, and behavioral dysregulation were consistently assigned higher importance in model predictions. These features likely reflect overlapping symptomatology captured by the DASS-21 and related self-reports, rather than independent drivers of mental health problems. Accordingly, the explainability results should be interpreted as illuminating model behavior and internal decision logic, not as evidence of causal mechanisms or independent risk factors. The prominence of symptom-related features underscores the tautological risk inherent in predicting mental health outcomes using variables that may themselves represent manifestations of psychological distress.

The findings highlight a substantial burden of psychological distress among university students during a period of social and political upheaval in Bangladesh, emphasizing the importance of accessible mental health support during times of crisis. While the explainable ML framework demonstrates potential value for pattern recognition and hypothesis generation, its application in this study is best understood as exploratory and sample-specific. Future research should prioritize probability-based sampling, longitudinal designs, and the

use of external validation cohorts to improve generalizability and temporal inference. Incorporating clinically validated diagnostic assessments and separating symptom indicators from contextual or antecedent factors would further strengthen model interpretability. Additionally, evaluating ML performance under more balanced prevalence conditions may yield more meaningful insights into predictive utility. In conclusion, this study provides a transparent, context-aware application of explainable ML to student mental health data in Bangladesh, offering insights into symptom patterns and model behavior while clearly delineating the methodological limits of inference. These findings may help guide future research directions and methodological considerations for mental health monitoring, particularly during periods of widespread social disruption.

Conclusions

This study applied explainable ML as an exploratory tool to examine depression, anxiety, and stress among Bangladeshi university students in a high-symptom, self-selected sample. Overall, psychological distress was highly prevalent, with substantial symptom overlap. ML models demonstrated modest predictive ability: SVM performed relatively better for depression and stress, while XGBoost showed more balanced classification for anxiety. Explainable analyses highlighted features related to sleep disturbance, cognitive complaints, and behavioral difficulties, reflecting overlapping symptomatology rather than independent causal predictors.

The contribution of this study is methodological and exploratory. We demonstrate the feasibility of applying explainable ML techniques for pattern recognition in student mental health data, provide a transparent framework for model interpretation, and establish benchmark performance metrics for future research. These findings underscore the importance of using explainable ML as a hypothesis-generating tool rather than a clinical prediction instrument. Future research should prioritize probability-based sampling, longitudinal designs, and the use of external validation cohorts to improve generalizability and temporal inference. Incorporating clinically validated diagnostic assessments and separating symptom indicators from contextual or antecedent factors would further strengthen model interpretability. In conclusion, this study provides a transparent, context-aware application of explainable ML to student mental health data in Bangladesh, offering insights into symptom patterns and model behavior while clearly delineating the methodological limits of inference.

Data availability

Data will be available upon a reasonable request to the corresponding author.

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Author contributions

Mst. Nilufar Yasmin (MNY) contributed to data processing, result interpretation, literature review, manuscript drafting, and editing. Mohammad Mizanur Rahman (MMR) performed data analysis and contributed to interpreting the results. Nusrat Y Ahmed (NYA), Muhammad Khairul Alam (MKA), Amrin Binte Ahmed (ABA) and Adisha Dulmini (AD) analyzed data and reviewed the manuscript. Nusrat Y Ahmed (NYA) and Atikur Rahman (AR) carried out critical manuscript review and editing. Rumana Rois (RR) conceptualized the research, designed the study and questionnaire, oversaw data collection and processing, supervised the entire project, and led study implementation. All authors have read and approved the final version of the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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